

Cutting SMS Lists

2026-01-09

```
library(dplyr)
library(ggplot2)
#library(cwi)
if (require(sf)) {
  sf::sf_use_s2(FALSE)
}
library(RSQLite)
library(DBI)
library(leaflet)
library(leaflet.extras2)
library(htmlwidgets)
library(tidygeocoder)
library(tidyverse)
library(data.table)
library(purrr)
library(leaflet)
library(lubridate)
library(reactable)
library(flextable)
library(magrittr)
library(dplyr)
library(randomizr)
```

Read in Data (from other directories)

```
setwd(r"(C:\Users\Ruairi\Documents\Coding Projects 2025\ct04_testing)")
vf_clean <- read.csv("clean_vf_with_history.csv")

scrubbed_phones_active_dems <- read.csv("scrubbed_numbers 2026-01-03 09-24-35.csv")
scrubbed_phones_active_dems$phone_number <- as.character(scrubbed_phones_active_dems$phone_number)
```

Merge Voterfile with Scrubbed Phones and Limit to Textable Numbers

```
vf_active_dems_good_phones <- left_join(vf_clean, scrubbed_phones_active_dems, join_by(phone == phone_n
  filter(fake_number != 'YES') %>%
  filter(phone_line_type == 'CELL PHONE') %>%
  filter(party_code %in% c('D') & status == 'A')
```

```
deduped_active_dems_good_phones <-
left_join(vf_active_dems_good_phones,
vf_active_dems_good_phones %>% group_by(phone) %>% summarise(phone_count=n()),
join_by(phone == phone)) %>% filter(phone_count == 1) #>% arrange(desc(phone_count), phone) #>% selec

# Only 702 phone numbers attributed to more than 1 person (1,444 people total)
# I drop them above
```

Random Assignment of Individuals to 8 Message Groups w/ Balance on Age and Gender

```
tier_1_towns <- c('New Haven', 'Hamden', 'West Haven', 'Milford', 'Stratford')

deduped_active_dems_good_phones <-
  deduped_active_dems_good_phones %>%
    mutate(tier1 = ifelse(town_name %in% tier_1_towns, 1, 0)) %>%
    mutate(age_bracket = cut(age,
                             breaks = c(18, 35, 50, 65, Inf),
                             labels = c("18-34", "35-49", "50-64", "65+"),
                             right = FALSE))

lower_tiers <- deduped_active_dems_good_phones %>% filter(tier1 == 0)

set.seed(2026)
lower_tiers$strata <- paste(lower_tiers$gender, lower_tiers$age_bracket, sep = "_")
lower_tiers$message <- block_ra(blocks = lower_tiers$strata,
                                conditions = c("local_choice", "local_voice", "local_dem", "local_soft",
                                                "dist_choice", "dist_voice", "dist_dem", "dist_soft"))
```

Balance checks

```
#table(lower_tiers$strata, lower_tiers$message)

table(lower_tiers$message, lower_tiers$age_bracket)
```

```
##
##      18-34 35-49 50-64 65+
## local_choice 1200  936  621 521
## local_voice  1202  936  620 521
## local_dem    1200  937  621 521
## local_soft   1200  937  619 520
## dist_choice  1203  936  619 522
## dist_voice   1200  937  620 521
## dist_dem     1202  936  619 521
## dist_soft    1200  936  620 521
```

```
table(lower_tiers$message, lower_tiers$gender)
```

```
##
##           F      M      U
## local_choice 2002 1225  51
## local_voice  2002 1226  51
## local_dem    2002 1226  51
## local_soft   2001 1224  51
## dist_choice  2002 1226  52
## dist_voice   2002 1225  51
## dist_dem     2002 1225  51
## dist_soft    2001 1224  52
```

```
table(lower_tiers$message, lower_tiers$town_name)
```

```
##
##           Ansonia Beacon Falls Bethany Branford Derby Durham East Haven
## local_choice      204           40      62      315      77      39      292
## local_voice       191           56      59      292      73      46      230
## local_dem         202           35      45      313      71      75      246
## local_soft        192           58      40      315      89      50      245
## dist_choice       195           52      45      332      77      43      275
## dist_voice        213           43      43      313      82      47      279
## dist_dem          193           48      60      304      68      50      247
## dist_soft         195           47      49      301      69      64      273
##
##           Guilford Middlefield Middletown Naugatuck North Branford
## local_choice      221           31      277      282           83
## local_voice       239           33      302      271           90
## local_dem         277           38      278      268           88
## local_soft        280           26      314      255           77
## dist_choice       241           26      263      287           98
## dist_voice        255           34      253      255           75
## dist_dem          281           43      251      263           83
## dist_soft         249           23      270      301           77
##
##           North Haven Orange Prospect Seymour Shelton Wallingford
## local_choice      185      139      53      126      234      341
## local_voice       182      131      63      142      249      350
## local_dem         185      117      53      145      276      311
## local_soft        184      120      57      128      230      348
## dist_choice       190      118      72      110      232      331
## dist_voice        199      119      57      137      257      348
## dist_dem          176      130      66      111      271      327
## dist_soft         178      141      54      123      261      320
##
##           Waterbury Woodbridge
## local_choice      175           102
## local_voice       187           93
## local_dem         156           100
## local_soft        163           105
## dist_choice       194           99
```

```
## dist_voice      182      87
## dist_dem        186     120
## dist_soft       185      97
```

```
counts <- lower_tiers %>%
  group_by(strata, message) %>%
  summarise(count = n(), .groups = "drop") %>%
  pivot_wider(names_from = message, values_from = count, values_fill = 0)

col_sums <- counts %>%
  select(-strata) %>%      # exclude the strata column
  summarise(across(everything(), sum)) %>%
  mutate(strata = "Total") # add a label for the total row

# Bind totals to the table
assignment_table_with_totals <- bind_rows(counts, col_sums)

assignment_table_with_totals %>% head(13)
```

```
## # A tibble: 13 x 9
##   strata local_choice local_voice local_dem local_soft dist_choice dist_voice
##   <chr>      <int>      <int>      <int>      <int>      <int>      <int>
## 1 F_18-34      762        763        762        762        763        762
## 2 F_35-49      575        575        575        575        575        575
## 3 F_50-64      371        370        371        371        370        371
## 4 F_65+        294        294        294        293        294        294
## 5 M_18-34      421        422        421        421        422        421
## 6 M_35-49      350        350        351        350        350        351
## 7 M_50-64      240        240        240        239        239        239
## 8 M_65+        214        214        214        214        215        214
## 9 U_18-34       17         17         17         17         18         17
## 10 U_35-49      11         11         11         12         11         11
## 11 U_50-64      10         10         10         9         10         10
## 12 U_65+        13         13         13         13         13         13
## 13 Total      3278      3279      3279      3276      3280      3278
## # i 2 more variables: dist_dem <int>, dist_soft <int>
```

```
lower_tiers %>% group_by(message) %>% summarise(avg_age = mean(age), med_age = median(age))
```

```
## # A tibble: 8 x 3
##   message      avg_age med_age
##   <fct>      <dbl>   <dbl>
## 1 local_choice  43.9     41
## 2 local_voice  44.0     41
## 3 local_dem    43.8     41
## 4 local_soft   43.8     41
## 5 dist_choice  43.9     41
## 6 dist_voice   43.9     41
## 7 dist_dem     43.8     40
## 8 dist_soft    43.9     41
```